MATEMATICI 2

Seminar 8

Ex.

V = sp. vect. cu prod. scalar

Să se arate că $\langle v, u \rangle = 0$, $\forall\, v \in V \Leftrightarrow u \in V$

$\Leftrightarrow u = 0$

Rez.

„$\Rightarrow$" Pt. $v = u \Rightarrow \langle u, u \rangle = 0 \Rightarrow u = 0$

$\langle v, 0 \rangle = \langle v, 0+0 \rangle = \langle v, 0 \rangle + \langle v, 0 \rangle = 2\langle v, 0 \rangle \Rightarrow$

$\Rightarrow \langle v, 0 \rangle = 0$

Ex. Fie V = sp. vect. real sau complex finit, cu prod. scalar

Să se arate că dacă $\langle v, u \rangle = \langle v, w \rangle$, $\forall v \in V \Leftrightarrow$

$\Leftrightarrow u = w$

„ $\Rightarrow$ " $\langle v, u \rangle = \langle v, w \rangle \Rightarrow$

$\Rightarrow \langle v, u \rangle - \langle v, w \rangle = 0$

$\langle u, v \rangle - \langle w, v \rangle = 0$

$\langle u - w, v \rangle = 0$

$\langle v, u - w \rangle = 0 \Rightarrow$

$\Rightarrow u - w = 0 \Rightarrow u = w$

„ $\Leftarrow$ " $u = w \Rightarrow \langle v, u \rangle = \langle v, w \rangle, \forall v \in V$

Ex. $f : \mathbb{R}_2[X] \to \mathbb{R}_2[X]$, $f(P) = P + P'$

Să se det. baze pt. $\ker f$, $\operatorname{Im} f$, $\sigma(f)$

Sp. de vectori proap.

Rez.

$$P(x) = ax^2 + bx + c$$

$$P'(x) = 2ax + b$$

$\operatorname{Ker} f = \{ p \in \mathbb{R}_2[x] \mid f(p) = 0 \}$

$f(p) = p + p' = ax^2 + (b+2a)x + (b+c) = 0$

$\forall x \in \mathbb{R} \Rightarrow$

$$\Rightarrow \begin{cases} a = 0 \\ b + 2a = 0 \Rightarrow b = 0 \\ b + c = 0 \Rightarrow c = 0 \end{cases}$$

$\Rightarrow \operatorname{Ker} f = \{0\} \Rightarrow \dim_{\mathbb{R}} \operatorname{Ker} f = 0 \Rightarrow$

$\Rightarrow \operatorname{Ker} f$ NU are bază

Met. 2 $\int_0^x$ $P = -P' \Rightarrow P = 0$

$\mathrm{Im} f = \{ f(p) \mid p \in \mathbb{R}_2[x] \}$

$f(p) = ax^2 + (b+2a)x + (b+c)$

$\dim \ker f + \dim \mathrm{Im} f = \dim \mathbb{R}_2[x] \Rightarrow$

$\dim \ker f = 0$

$\Rightarrow \dim \operatorname{Im} f = \dim \mathbb{R}_2[X]$

$\operatorname{Im} f \subset \mathbb{R}_2[X]$

$\Rightarrow \operatorname{Im} f = \mathbb{R}_2[X]$

$B_{\operatorname{Im} f} = \{1, X, X^2\}$

Fie $\lambda \in \sigma(f) \Rightarrow \exists\, p \neq 0$ a.î. $f(p) = \lambda p \Rightarrow$

$\Rightarrow p + p' = \lambda p \Rightarrow (\lambda - 1)p = p'$

Dc. $\lambda - 1 \neq 0 \Rightarrow \text{grad } p = \text{grad } p'$ fb $\Rightarrow$

$\Rightarrow \lambda - 1 = 0 \Rightarrow \lambda = 1 \Rightarrow$

$\Rightarrow \sigma(f) = \{1\}$

$$V_{\lambda_1} = \{ p \in \mathbb{R}_2[X] \mid f(p) = \lambda_1 \cdot p \}$$

$$f(p) = p + p' = p \Rightarrow p' = 0 \Rightarrow 2ax + b = 0, \forall x \in \mathbb{R} \Rightarrow$$

$$\Rightarrow \begin{cases} 2a = 0 \Rightarrow a = 0 \\ b = 0 \end{cases} \Rightarrow p = c \in \mathbb{R}$$

$$V_{\lambda_1} = \{ c \mid c \in \mathbb{R} \}$$

$$\mathcal{B}_{V_{\lambda_1}} = \{1\}$$

Consecință Lagrange

$f : [a, x] \to \mathbb{R}$

$$\frac{f(x) - f(a)}{x - a} = f'(c) \Rightarrow f(x) = f(a) + t$$

Ex. Fie $f : \mathbb{C}^2 \to \mathbb{C}^2$, $f(z_1, z_2) = (\alpha z_1 + \beta z_2, z_1 - \beta z_2)$

$\alpha, \beta = ?$ a.î. f bij.

obs. Ex. Dc. $f : V \to V$, V finit, f liniară $\Rightarrow$ atunci f inj. $\Leftrightarrow$ f surj.

$\alpha, \beta = ?$ a.î. f inj. $\Leftrightarrow \alpha, \beta = ?$ a.î. $\ker f = \{0\}$?

$\mathbb{C}^2$ sp. vectorial peste $\mathbb{C}$

$$f(z_1, z_2) = f(z_3, z_4)$$

$$(\alpha z_1 + \beta z_2, \alpha z_1 - \beta z_2) = (\alpha z_3 + \beta z_4, \alpha z_3 - \beta z_4)$$

$$(\alpha z_1 + \beta z_2 - \alpha z_3 - \beta z_4, \alpha z_1 - \beta z_2 - \alpha z_3 + \beta z_4) = (0, 0)$$

$$(\alpha(z_1 - z_3) + \beta(z_2 - z_4), \alpha(z_1 - z_3) - \beta(z_2 - z_4)) = (0, 0)$$

$$\Rightarrow \alpha(z_1 - z_3) + \beta(z_2 - z_4) = 0$$

$$\alpha(z_1 - z_3) - \beta(z_2 - z_4) = 0 \quad (+)$$

$$2\alpha(z_1 - z_3) = 0$$